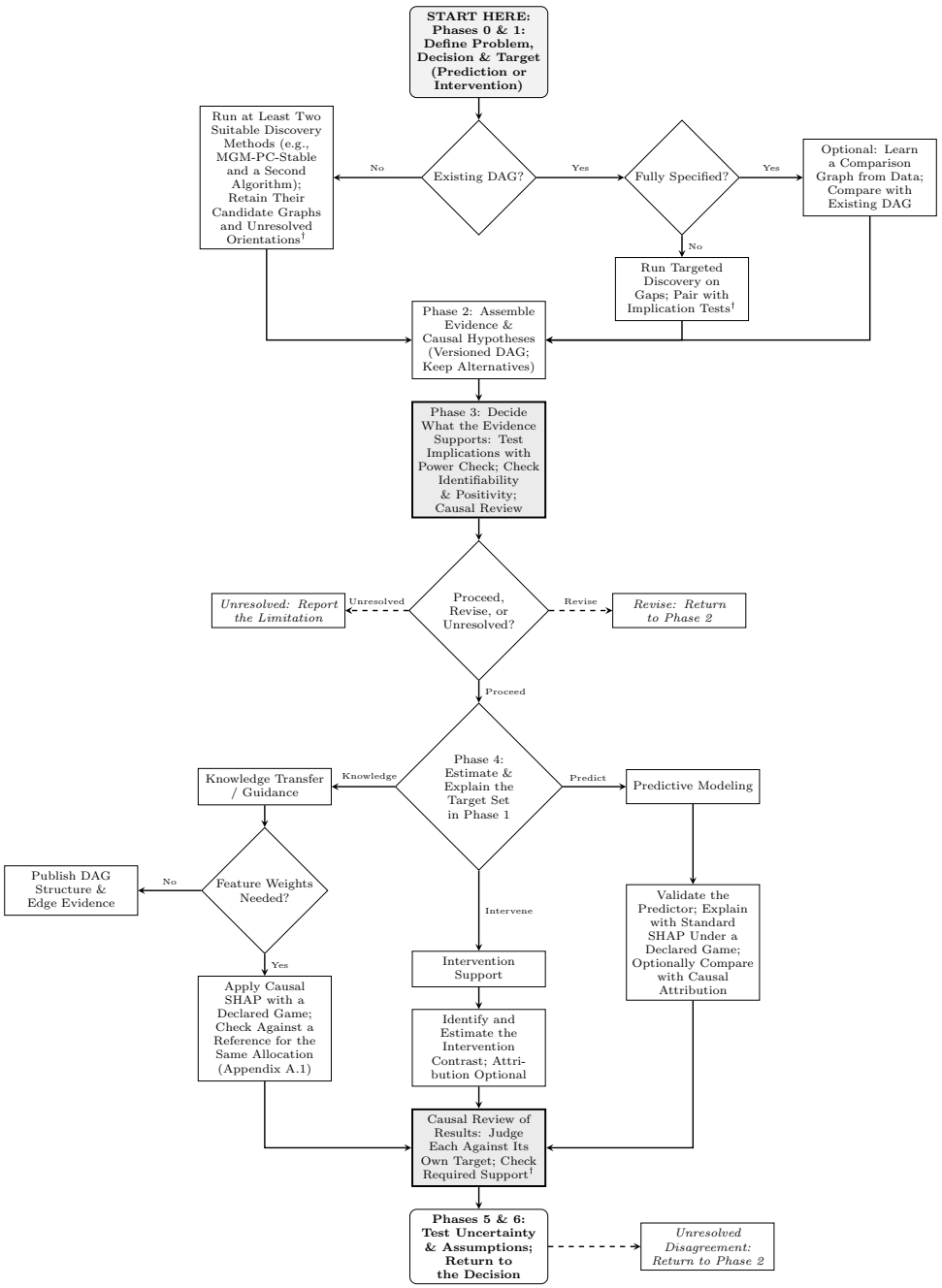
Playbook: DAG Construction, Refinement, and Causal Attribution

Phase-Based Decision Guide for Causal Modeling Pipeline

October 2026 (revised 8 October)

Scope & Prerequisites

Scope: Assumes availability of a labeled dataset suitable for validation. Dataset identification and raw ETL are out of scope. Prediction-only analyses may proceed directly to Predictive Modeling.
Core principle: No single method establishes a DAG or its attributions. Every step pairs a method with a second method that answers the same question differently, or with a known quantity, and every branch ends in causal review.



Shaded boxes with heavy borders are causal review steps. †Recommendation based on reasoning; not tested in the accompanying simulation study. Phase numbers are local to this schematic and differ from manuscript Steps 0–6.

Key Execution Rules & Safeguards

1. **Never Rely on One Method:** Pair each method with a second method that answers the same question differently, or check it against a known quantity, such as a simulated ground truth or an effect established in prior studies. Agreement can raise confidence when the methods have different sources of error; agreement alone does not validate the DAG, because methods run on the same data share its selection and measurement error and can succeed for a shared artifact of the data [1, 2]. PC estimates a Markov equivalence class under its assumptions, so retain unresolved orientations and alternative candidate graphs, and treat intersecting outputs as a stated precision-over-recall choice rather than a default [3]; use tests suited to the variable types [4]. Disagreement goes to causal review rather than being settled by either method alone.
2. **Represent Selection Explicitly:** In a selected cohort, some implications of a correct DAG fail because of how the cohort was selected, not because the DAG is wrong. Draw a selection indicator with arrows from its causes, and treat the analyzed cohort as conditioned on it, before testing implications [5].
3. **Check Power Before Trusting a Pass:** Implication tests and discovery may have low power for rare exposures, especially in small samples [6]. Rejection challenges the graph or the testing assumptions; non-rejection does not confirm the graph or support a present edge. Account for graph selection when reusing discovery data for inference [7].
4. **Sparse Nodes Limit Attribution Too:** Rare exposures can make attribution estimates unstable. Check support for the chosen game; distinguish non-identification from finite-sample bias or imprecision, and do not infer unimportance from either [8].
5. **Ordering Alone Does Not Propagate Interventions:** In our matched full-DAG comparison, restricting feature orders produced no detected improvement in intervention-ranking agreement. The separate prototype using known mechanisms showed higher agreement, provisionally; its order law and computational budget also differed. Restricting players to assumed ancestors excludes non-ancestors by construction [9–11].
6. **Causal Attribution Is Only as Good as Its Inputs:** Causal attribution methods use method-specific causal information, such as order constraints or graph structure, with mechanisms where required [9–11]. Errors in those inputs carry into the attributions. Report sensitivity to the retained alternative graphs.[†]
7. **Look for Positivity Gaps in the Target's Support:** If the contrast requires exposure within a stratum where it cannot occur, such as a drug never used in one era, it is not identified from those data without additional assumptions [8]. Inspect the support the target requires before trusting its attribution; depth itself is not the criterion.
8. **Causal Review Closes Every Branch:** A clinical team reviews structure and attributions together; unresolved disagreement returns the analysis to the DAG.[†]

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